University of Cape Town



Prescriptive Analytics Assignment 1

Optimal Mobile Generator Distribution for Telecommunication Network

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Optimal Site Generator Distribution

Data Sources: Vodacom data, localized for the Vodacom KZN Region

Problem Statement * will add more

Load shedding and insecure power supply put a strain on network servicing sites, affecting coverage and availability of those sites with material impact on site revenue generation from user traffic flows. Running generators at sites impacted by power disruptions offers the immediate solution to ensuring sites continue to run during interrupted power supply; however, there is a limited supply of generators that can be deployed to sites, as well as the cost associated with the deployment and running of the generators. Vodacom would like to develop a strategy for the optimal distribution of generators to important sites with the minimum cost attached whilst ensuring coverage needs and availability are maintained.

Key Definitions:

- Atoll Site ID: Identifier for a physical structure that houses antennas, transmitters, receivers, and networking hardware needed to transmit and receive voice, data, and internet signals. These have a set coverage shape that defines the span of the site's coverage over a physical space (word this better and added refrence)
- Site Availability: The percentage of time that a specific telecom site, including its radio equipment, power supplies, backhaul, and virtual or physical connection links, is fully operational and capable of processing traffic. (https://medium.com/@saeedfelegari/site-availability-in-telecom-metrics-architecture-and-the-real-requirements-behind-the-numbers-37cf574ba711.
  - This metric is measured as the percentage number of seconds a site is operational over the total number of seconds in a month, with 100%= full availability.

EDA: Site Counts and Spatial Distributions

Section 1: Deterministic MLP Formulation

The generator deployment problem lends itself to the set covering problem formulation. In this case, we would like to deploy the minimum number of sites

(and the associated cost) subject to the constraint that the aggregate availability over the region is >=95%. Sites also neighbour each other such that their coverage may overlap in certain regions where sites are densely clustered.

Assumptions:

- Each site can get only one generator
- Average site coverage= 30K radius
- Model for a Stage 1 load shedding schedule with at most 1 power outage per day.

What I need: Coverage matrix to get the distances between sites.

LP Formulation:

Mathematical formulation

Sets and parameters

Let J be the set of candidate generator sites and I the set of elementary demand cells obtained by overlaying the coverage polygons (or from a regular grid over the region of interest).

$$a_{ij} = \begin{cases} 1 & \text{if cell } i \in I \text{ lies within the coverage polygon of site } j \in J, \\ 0 & \text{otherwise,} \end{cases}$$

where c_j is the cost of placing a generator at site j ($c_j = 1$ if the objective is simply the number of generators), w_i is the weight of cell i (area, or population/traffic if available), and $\alpha \in (0, 1]$ is the minimum required fraction of weighted coverage.

Decision variables

$$x_j = \begin{cases} 1 & \text{if a generator is placed at site } j, \\ 0 & \text{otherwise,} \end{cases} \qquad y_i = \begin{cases} 1 & \text{if cell } i \text{ is covered,} \\ 0 & \text{otherwise.} \end{cases}$$

Model A: minimum-cost partial set covering

$$
\begin{aligned}
\min \quad & \sum_{j \in J} c_j \, x_j \\
\text{s.t.} \quad & y_i \le \sum_{j \in J} a_{ij} \, x_j && \forall \, i \in I, \\
& \sum_{i \in I} w_i \, y_i \;\ge\; \alpha \sum_{i \in I} w_i, \\
& x_j \in \{0,1\} && \forall \, j \in J, \\
& y_i \in \{0,1\} && \forall \, i \in I.
\end{aligned}
$$

The first constraint links coverage to placement: a cell can only be counted as covered if at least one selected site's polygon contains it. The second enforces the minimum weighted coverage target. Setting $\alpha = 1$ and dropping the y_i variables recovers the classical set covering problem $\sum_j a_{ij} x_j \ge 1$ for all i.

Model B: maximal covering (budget- or count-constrained)

$$
\begin{aligned}
\max \quad & \sum_{i \in I} w_i \, y_i \\
\text{s.t.} \quad & y_i \le \sum_{j \in J} a_{ij} \, x_j && \forall \, i \in I, \\
& \sum_{j \in J} c_j \, x_j \le B, \\
& x_j \in \{0,1\} && \forall \, j \in J, \\
& y_i \in \{0,1\} && \forall \, i \in I,
\end{aligned}
$$

where B is the available budget. With $c_j = 1$, replace the budget constraint by $\sum_j x_j = p$ to place exactly p generators.

OR :

$$
\begin{aligned}
\min \quad & Z = \sum_{j \in J} g_j x_j \\
\text{subject to} \quad & \sum_{k \in K} w_k y_k \geq \alpha \sum_{k \in K} w_k \\
& y_k \leq \sum_{j \in J} C_{kj} x_j, && \forall k \in K \\
& \sum_{j \in J} x_j \leq G \\
& x_j \in \{0,1\}, && \forall j \in J \\
& y_k \in \{0,1\}, && \forall k \in K
\end{aligned}
$$

Notation

- (J): set of candidate generator sites.

- (K): set of geographical coverage cells.

- (g_j): cost of deploying a generator at site (j).

- (x_j): 1 if a generator is deployed at site (j), and 0 otherwise.

- (y_k): 1 if geographical cell (k) retains coverage during load shedding, and 0 otherwise.

- (C_{kj}): 1 if site (j) can adequately cover cell (k), and 0 otherwise.

- (w_k): weight of cell (k), such as its area

. - ( ): minimum required coverage proportion (e.g. 0.95). - (G): maximum number of available generators.

Section 2: Deterministic MOLP Formulation-Revenue and Site Rankings

Site importance and multi-objective extension

Additional parameters

Let R_i and T_i denote the revenue and traffic of network site $i \in I$. Normalise each KPI to $[0, 1]$:

$$\tilde{R}_i = \frac{R_i - \min_k R_k}{\max_k R_k - \min_k R_k}, \qquad \tilde{T}_i = \frac{T_i - \min_k T_k}{\max_k T_k - \min_k T_k}.$$

The composite importance score of site i is

$$s_i = \theta\, \tilde{R}_i + (1-\theta)\, \tilde{T}_i, \qquad \theta \in [0,1],$$

where θ sets the relative emphasis on revenue versus traffic. Alternatively, if only the ordinal rank $\rho_i \in \{1, \dots, N\}$ is used (1 = highest revenue), a rank-based weight is

$$s_i = \frac{N - \rho_i + 1}{N} \qquad \text{or} \qquad s_i = \frac{1}{\rho_i}.$$

The total importance of the network is $S = \sum_{i \in I} s_i$. Sites with s_i in the top tier are collected in the set $I^{\text{top}} \subseteq I$ (for example, the top 10% by revenue rank).

Model C: weighted-sum objective

Using the same decision variables x_j (generator placed at site j) and y_i (site i is covered) as before, let $C = \sum_{j \in J} c_j$ be the cost of deploying at every candidate site (used for scaling). Then

$$\begin{aligned}
\min \quad & (1-\lambda)\, \frac{1}{C} \sum_{j \in J} c_j\, x_j \;-\; \lambda\, \frac{1}{S} \sum_{i \in I} s_i\, y_i \\
\text{s.t.} \quad & y_i \le \sum_{j \in J} a_{ij}\, x_j && \forall\, i \in I, \\
& \sum_{i \in I} w_i\, y_i \;\ge\; \alpha \sum_{i \in I} w_i, \\
& y_i = 1 && \forall\, i \in I^{\text{top}} \quad \text{(optional)}, \\
& x_j \in \{0,1\},\; y_i \in \{0,1\},
\end{aligned}$$

where $\lambda \in [0,1]$ trades off cost against protected importance. Both terms are scaled to $[0,1]$ so that λ has a meaningful interpretation. At $\lambda = 0$ the model reduces to Model A.

Model D: ε-constraint (cost minimisation with an importance floor)

$$\begin{aligned}
\min \quad & \sum_{j \in J} c_j \, x_j \\
\text{s.t.} \quad & y_i \le \sum_{j \in J} a_{ij} \, x_j && \forall \, i \in I, \\
& \sum_{i \in I} w_i \, y_i \;\ge\; \alpha \sum_{i \in I} w_i, \\
& \sum_{i \in I} s_i \, y_i \;\ge\; \beta \, S, \\
& x_j \in \{0,1\}, \; y_i \in \{0,1\},
\end{aligned}$$

where $\beta \in [0,1]$ is the minimum fraction of total weighted revenue/traffic that must be protected. This is equivalent to Model C but gives the decision-maker a directly interpretable target. Varying β (and α) traces the Pareto frontier of cost against protected importance.

Model E: lexicographic (priority-ordered) objectives

Stage 1 solves Model A to obtain the minimum cost z_1^*. Stage 2 fixes cost at (or within a tolerance δ of) the optimum and maximises protected importance:

$$\begin{aligned}
\max \quad & \sum_{i \in I} s_i \, y_i \\
\text{s.t.} \quad & \sum_{j \in J} c_j \, x_j \le (1+\delta) \, z_1^*, \\
& \text{all constraints of Model A.}
\end{aligned}$$

This guarantees that, among all cost-minimal (or near cost-minimal) deployments, the one protecting the most important sites is chosen.

Optional: favouring important generator locations

If you also want the *chosen generator sites themselves* to be high-ranked, add a term in the site-selection variables, with s_j the importance score of candidate site j:

$$\max \quad \sum_{i \in I} s_i \, y_i \;+\; \gamma \sum_{j \in J} s_j \, x_j, \qquad \gamma \ge 0.$$

Option F

$$
\begin{aligned}
\max \quad & Z = \lambda_R \sum_{j \in J} \tilde{r}_j x_j + \lambda_S \sum_{j \in J} s_j x_j - \lambda_C \sum_{j \in J} \tilde{g}_j x_j \\
\text{subject to} \quad & \sum_{k \in K} w_k y_k \geq \alpha \sum_{k \in K} w_k \\
& y_k \leq \sum_{j \in J} C_{kj} x_j, && \forall k \in K \\
& \sum_{j \in J} x_j \leq G \\
& x_j \in \{0,1\}, && \forall j \in J \\
& y_k \in \{0,1\}, && \forall k \in K
\end{aligned}
$$

where

- J: Set of candidate sites where generators could be deployed.
- K: Set of geographic cells used to represent the coverage area.
- $x_j \in \{0,1\}$: Binary decision variable equal to 1 if a generator is deployed at site j, and 0 otherwise.
- $y_k \in \{0,1\}$: Binary indicator equal to 1 if geographic cell k retains adequate network coverage during load shedding, and 0 otherwise.
- $C_{kj} \in \{0,1\}$: Coverage indicator equal to 1 if site j, when operational, provides adequate coverage to cell k, and 0 otherwise.
- w_k: Area weight of geographic cell k.
- α: Minimum required proportion of the total geographic area that must retain adequate coverage.
- G: Maximum number of generators available for deployment.
- $\tilde{r}_j$: Normalised revenue associated with site j.
- s_j: Normalised importance score for site j, derived from its business ranking.
- $\tilde{g}_j$: Normalised generator deployment cost at site j.
- $\lambda_R, \lambda_S, \lambda_C \geq 0$: Objective weights representing the relative importance of revenue, site ranking, and generator cost, respectively.

Section 3: Simulated Annealing Formulation*

Let the candidate solution be represented by a binary vector:

Simulated Annealing Formulation

Let the candidate solution be represented by a binary vector:

$$\mathbf{x} = (x_1, x_2, \dots, x_n), \qquad x_j \in \{0, 1\}, \quad j = 1, \dots, n$$

where (x_j = 1) indicates that a generator is deployed at site (j), and (x_j = 0) otherwise.

The initial solution is denoted by

$$\mathbf{x^0} = (x_1, x_2, \dots, x_n)$$

At each iteration (t), a neighbouring solutio is generated by modifying the current solution. For example,

$$\mathbf{x}^{(t)} = (1, 0, 0, 1, 1)$$

could generate the neighbouring solution

$$\mathbf{x}' = (0, 0, 0, 1, 1)$$

The objective function is evaluated at the current and neighbouring solutions. For the initial set-covering problem, the objective is to minimise the total number of generators deployed:

$$\min_{\mathbf{x}} \quad f(\mathbf{x}) = \sum_{j=1}^{n} x_j$$

The simulated annealing algorithm uses the change in objective value,

$$\Delta f = f(\mathbf{x}') - f\left(\mathbf{x}^{(t)}\right)$$

to determine whether the neighbouring solution should be accepted, with the acceptance decision also depending on the current temperature.

Discretizing the Problem: Stochastic Load Shedding Scheduling

What is different:
One of my key assumptions initial LP formulation is that the load shedding schedule is deterministic and that all sites are experiance power outages at the same time. In reality this is not the case. There is a stochastic element to the prevalence of the power outages and as such not all sites will experiance interrupted power at the same time and require generators. In light of this I want to model the problem as a discrete event and run simulations based on the appropriate statistical distributions of the loadshedding event.

Time, zones and scenarios

Let $[0, H]$ be the planning horizon (e.g. one month), discretised into periods $t \in T = \{1, \dots, |T|\}$ of length Δ hours. Each network site $i \in I$ belongs to exactly one load-shedding zone (block) $g(i) \in G$. Let $\Omega = \{1, \dots, N\}$ be a set of independent simulation replications (scenarios), each with probability $1/N$.

Stochastic outage process

Load-shedding stage. Let $\sigma(t) \in \{0, 1, \dots, 8\}$ be the national stage, modelled as a discrete-time Markov chain with transition matrix $P = [p_{kl}]$,

$$p_{kl} = \Pr\left(\sigma(t+1) = l \mid \sigma(t) = k\right).$$

Zone outages. Zone g is scheduled to be off when stage $\sigma(t)$ includes its block in the rotation. Let $u_g(t) \in \{0, 1\}$ be the zone outage indicator. Realised outage episodes deviate from the schedule, so for the m-th episode in zone g:

$$\begin{aligned} A_{g,m} &\sim F_A(\cdot \mid \sigma,\, \phi_A) \quad \text{(time between episodes)}, \\ D_{g,m} &\sim F_D(\cdot \mid \sigma,\, \phi_D) \quad \text{(episode duration)}, \end{aligned}$$

where, for example, $A \sim \text{Weibull}(k_A, \lambda_A)$ or $\text{Exponential}(\mu_\sigma)$, and $D \sim \text{Lognormal}(\mu_D, \sigma_D^2)$ or $\text{Gamma}(k_D, \theta_D)$, with parameters estimated from historical outage logs (maximum likelihood, with goodness-of-fit checked by AIC and K–S tests). Episode start and end times are

$$\tau^{\text{start}}_{g,m} = \tau^{\text{end}}_{g,m-1} + A_{g,m}, \qquad \tau^{\text{end}}_{g,m} = \tau^{\text{start}}_{g,m} + D_{g,m}.$$

Site-specific faults. Independently of load shedding, site i experiences local

failures (cable theft, breaker trips, etc.) as a Poisson process with rate ν_i and repair time $D_i^{\text{loc}} \sim F_{\text{loc}}$, giving indicator $v_i(t)$.

Site outage indicator. Site i is without grid power in period t of scenario ω if either its zone or the site itself is down:

$$o_{it}^{\omega} = 1 - (1 - u_{g(i)}^{\omega}(t))(1 - v_i^{\omega}(t)) \in \{0, 1\}.$$

Correlation between sites is therefore induced through the shared zone process u_g and the shared stage $\sigma(t)$, while sites in different zones are only coupled through $\sigma(t)$.

Discrete-event simulation specification

- **State:** $\mathbf{S}(t) = (\sigma(t), \{u_g(t)\}_{g \in G}, \{v_i(t)\}_{i \in I})$, optionally extended with generator fuel level $f_j(t)$.
- **Event types:** stage change; zone outage start; zone outage end; local fault start; local fault end; (optional) generator fuel exhaustion and refuel.
- **Event list:** a priority queue ordered by event time. The next event $e^* = \arg\min_e \tau_e$ is popped, the clock advances to τ_{e^*}, the state is updated and new events are scheduled by sampling from F_A, F_D and F_{loc}.
- **Output:** for each replication ω, the outage indicators o_{it}^{ω} aggregated on the grid T.

Scenario-based outage metrics

The outage duration (hours) of site i in scenario ω is

$$d_i^{\omega} = \Delta \sum_{t \in T} o_{it}^{\omega}.$$

With R_i the revenue rate of site i (per hour) and s_i its importance score, the weighted outage exposure in scenario ω is

$$E^{\omega} = \sum_{i \in I} \sum_{t \in T} \Delta \, s_i \, o_{it}^{\omega}.$$

Decision variables

$$
\begin{aligned}
x_j &\in \{0,1\} && \text{generator placed at candidate site } j \quad \text{(first stage)},\\
z^{\omega}_{ijt} &\in \{0,1\} && \text{generator } j \text{ supplies site } i \text{ in period } t,\ \text{scenario } \omega \quad \text{(recourse)},\\
y^{\omega}_{it} &\in [0,1] && \text{site } i \text{ is backed up in period } t,\ \text{scenario } \omega.
\end{aligned}
$$

Recourse constraints (hold for every $\omega \in \Omega$)

$$
\begin{aligned}
z^{\omega}_{ijt} &\le a_{ij}\, x_j && \forall\, i,j,t,\\
\sum_{j\in J} z^{\omega}_{ijt} = y^{\omega}_{it} &\le o^{\omega}_{it} && \forall\, i,t,\\
\sum_{i\in I} l_i\, z^{\omega}_{ijt} &\le K_j\, x_j && \forall\, j,t,
\end{aligned}
$$

where l_i is the electrical load of site i and K_j the capacity of the generator at j. A site can only be supplied by a placed generator whose polygon contains it, only while it is actually in outage, and only up to the generator's capacity.

Model F: expected-value SAA (minimum cost with expected protection target)

$$
\begin{aligned}
\min \quad & \sum_{j\in J} c_j\, x_j\\
\text{s.t.} \quad & \frac{1}{N}\sum_{\omega\in\Omega}\sum_{i\in I}\sum_{t\in T} \Delta\, s_i\, y^{\omega}_{it} \;\ge\; \beta\,\frac{1}{N}\sum_{\omega\in\Omega} E^{\omega},\\
& \text{recourse constraints for all } \omega \in \Omega,\\
& x_j \in \{0,1\}.
\end{aligned}
$$

The constraint requires that, on average, at least a fraction β of the importance-weighted outage exposure is covered by generators.

Model G: chance-constrained version (reliability across scenarios)

Introduce $q^\omega \in \{0,1\}$ equal to 1 if scenario ω meets the protection target:

$$\begin{aligned}
\min \quad & \sum_{j\in J} c_j\, x_j \\
\text{s.t.} \quad & \sum_{i\in I}\sum_{t\in T} \Delta\, s_i\, y_{it}^{\omega} \;\geq\; \beta\, E^{\omega} - M^{\omega}\,(1-q^{\omega}) \quad \forall\, \omega \in \Omega, \\
& \frac{1}{N}\sum_{\omega\in\Omega} q^{\omega} \;\geq\; 1-\eta, \\
& \text{recourse constraints for all } \omega \in \Omega, \\
& x_j \in \{0,1\},\; q^{\omega} \in \{0,1\},
\end{aligned}$$

with $M^\omega = \beta E^\omega$ a valid big-M, and $\eta \in (0,1)$ the tolerated probability of missing the target. Equivalently,

$$\Pr\left(\sum_{i,t} \Delta\, s_i\, y_{it} \geq \beta E\right) \geq 1-\eta.$$

Model H: cost–shortfall trade-off with risk measure

Let the unserved weighted exposure be $L^\omega = E^\omega - \sum_{i,t} \Delta\, s_i\, y_{it}^{\omega} \geq 0$. Then

$$\min \;\sum_{j\in J} c_j\, x_j \;+\; \kappa\Big[(1-\lambda)\,\mathbb{E}[L] \;+\; \lambda\,\mathrm{CVaR}_\alpha(L)\Big],$$

where

$$\mathrm{CVaR}_\alpha(L) = \min_{\zeta\in\mathbb{R}} \;\zeta + \frac{1}{(1-\alpha)N}\sum_{\omega\in\Omega} \big(L^\omega - \zeta\big)^+,$$

and κ converts unserved exposure into monetary units. This penalises the bad-outage tail rather than just the average.

Simulation–optimisation loop and sample size

1. Fit F_A, F_D, P, F_{loc} to historical outage data.
2. Simulate N replications and compute o_{it}^{ω}.
3. Solve the SAA model (F, G or H) to obtain x^*.
4. **Out-of-sample validation:** simulate $N' \gg N$ fresh replications with x^*

fixed and estimate

$$\widehat{\mu} = \frac{1}{N'} \sum_{\omega'} \Phi(x^*, \omega'), \qquad \widehat{\mu} \pm z_{1-\gamma/2} \, \frac{\widehat{s}}{\sqrt{N'}},$$

where $\Phi(x, \omega)$ is the protected (or unserved) exposure under placement x in scenario ω.

5. Increase N until x^* and the optimality gap stabilise.